SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. TEAM AND PROJECT OVERVIEW

Team Name

JJ

Project / Solution Name

ESG-Aware Receivables Risk & Collection Prioritization for Malaysian SMEs

Name (Leader)

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Email & Phone (Leader)

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University Affiliation(s)

xiamen university malaysia

Live Prototype URL

https://esg-ai-advisor.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/Lyuqiuyao/sdc-hackathon-JJ

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Arus is a three-tier web application deployed on Replit. Its React 19 + TypeScript single-page admin dashboard, built with Vite, Tailwind CSS and shadcn/ui, provides ten modules: Dashboard, Accounts Receivable, Customers, ESG Documents, Risk Triggers, AI Collection Planning, Approval Centre, Execution Results, Audit Logs and Settings. It consumes typed OpenAPI contracts via a TanStack Query client.

The Express 5 API validates requests with Zod, runs scoring, queue sorting and integer monetary logic, and persists typed workspace data into PostgreSQL JSONB through Drizzle. Row locks secure concurrent approvals, with audit logs attached to every commit. AI tasks run outside transactions: server-side OpenAI gpt-5-mini generates English/Malay bilingual outputs under strict JSON Schema for routing, extraction and profiling, with a 45-second non-retry timeout. Local validation and rule-based fallbacks prevent workflow stalls on failures. Approved collection notices go out via Nodemailer SMTP; unapproved actions are logged as simulations.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9 Node.js 24 esbuild 0.28	Implementation language, server runtime, API bundling, structured logging	Apache-2.0 MIT
React 19 Vite 7 Tailwind CSS 4 shadcn/ui（Radix UI）	Admin-side single-page application, build pipeline, styling tokens, accessible components	MIT
TanStack Query 5 Wouter 3 Recharts 2.15	Server - state caching, routing, KPI and risk charts	MIT
Express 5.2 Zod 4 Orval 8	REST API, request/response validation, OpenAPI + Zod code generation	MIT
Drizzle ORM 0.44 pg 8.23	Typed PostgreSQL JSONB store, transactions, row locks, atomic audit	Apache-2.0 MIT
OpenAI SDK 7 Nodemailer 10	gpt - 5 - mini structured outputs; SMTP delivery of approved collection notices	Apache-2.0 MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAI Responses API — gpt-5-mini (integrated via Replit AI)	Company routing, letter extraction, ESG profiling, bilingual English/Malay notice drafting	Sponsor-provided (Replit-hosted)
SMTP relay via Nodemailer (optional, standards-compliant provider)	Send approved collection notices; log as simulated execution when SMTP is unavailable	Free

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit — Cloud Workspace & Autoscale Deployment	Development, hosting, API routing, build / workflow execution, secret management	Replit Core / Autoscale
Replit AI Integration — Managed OpenAI Access	Server - side gpt - 5-mini calls using platform - managed credentials; no private API keys	Sponsor - provided integration
Replit - hosted PostgreSQL (DATABASE_URL)	Typed JSONB workspaces, proposal versions, approvals, executions and audit logs	Managed PostgreSQL

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
OpenAI gpt-5-mini	OpenAI, via Replit AI Integration	Pre-trained; prompt-only structured output, no fine-tuning or autonomous training	OpenAI API Terms / Replit Terms
Deterministic Rule Engine (non-machine-learning model)	In-house team-built TypeScript	In-house scoring, queue ranking, integer amount arithmetic, template fallback	Project source code (owned by the team)

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Arus demo workspace	Team-generated seed data	Synthetic data	Team-created	No

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit Agent (Claude Sonnet 4.6 / Sonnet 5)	Workspace scaffolding, React pages, Express routes, schema, tests, document drafts	Heavy
DeepSeek-V3.2 chat assistant	Code review, debugging, refactoring suggestions, wording and documentation	Medium

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Before committing, we reviewed every AI-generated change. We ran TypeScript type checks , OpenAPI / Zod contract checks, seed workflow tests, and browser walkthroughs of all interfaces — including deliberate AI failure runs — to confirm that rule fallbacks still complete the workflow. AI notifications are drafts only: admins must edit and approve them, and mock mode is enabled by default. Credentials are stored in Replit Secrets and redacted from logs. The team is responsible for every line of committed code, tests, and dependency declarations.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit / OpenAI	Replit AI integration — hosted gpt-5-mini access, Replit hosting and PostgreSQL	Prototyping hosting, hosted database, and server-side AI calls without private keys

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to delivery: the prototype is hosted on Replit Autoscale, using a hosted PostgreSQL and platform-hosted AI credentials, so no private API keys are needed and the demo runs for judges without setup. In principle it is replaceable: the stack is standard Node.js and PostgreSQL, and when AI is unavailable the workflow degrades to deterministic rules, so only AI enhancements and draft quality are lost. Replacing it would mean requiring new hosting, a database, and standalone AI credentials.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo workspace contains only synthetic company, invoice, ESG, and correspondence records. No real customer or personal data is collected; shared demo roles have no production credentials, so real data must never be imported. Uploaded letters are stored as text in the PostgreSQL workspace; recipient addresses are used only for that single send. AI requests send minimal synthetic context to the platform-hosted OpenAI endpoint with store=false, and document text is treated as untrusted data rather than instructions. Credentials are stored in Replit Secrets and never appear in the repository or UI. Audit logs record actions and state changes, not message bodies. Data is retained until an admin resets the workspace.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination or misattribution: gpt-5-mini output is constrained by a strict JSON Schema and validated locally; extracted values must map to source text lines, otherwise the response is discarded, rules take over, and ambiguous emails are never auto-archived. Bias: the ESG weighted queue uses fixed, visible weights that employees can override. Provider dependency : AI requires the Replit AI integration; a 45-second timeout, zero retries, and deterministic fallbacks keep the workflow available, and the retired Gemini path is rejected. Other limitations: English / Malay templates only, no OCR for scanned PDFs, input size limits, a single shared demo workspace, and no production identity controls.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☑ The prototype described here is accessible to judges through the links provided in Section 1.

☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

LYU QIUYAO

[Date]

2026.10.24